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Sámi Ethnoentomology and the More-than-Musical Aesthetic of Mosquitoes

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ABSTRACT

Across human history, few sounds in nature have attracted as much hostility as the mosquito's high-pitched whine. Despite occupying a very marginal position in multispecies scholarship – with an even smaller and rather ambitious body of literature addressing questions of musicality or even simply of aesthetic values – it is almost ironic that in entomology the very behavior that enables the reproduction of this infamous dipteran is routinely imagined in explicitly musical terms: the *dancing swarm*. Taking that notion as a point of departure, this article asks: how might mosquito sound and movement be taken seriously as aesthetic practices in their own right? And what forms of listening are required to "loosen" the encultured habits that consign this insect's sound to the status of mere nuisance? To address these questions, we move away from ethno-anthropocentric analogies and dominant epistemologies, and instead attend to knowledge systems that—much like the insect itself—have been positioned at the margins of global listening. Across the Arctic, diverse Indigenous traditions of multispecies storytelling frame the mosquito as kin rather than foe, opening a space for science to hear what buzzing may disclose in a rapidly changing world: from the insect's role in fragile ecosystems to the consequences of biodiversity loss, and, at times, to unexpected aesthetic registers of value. The article focuses on Sámi multispecies storytelling, where joik—understood here as biocultural heritage and more-than-musical expression—offers a distinctive lens for engaging mosquito sounds. By situating the insect as an active contributor to this intangible heritage, Sámi understandings of joik unsettle anthropocentric assumptions about mosquito musicality and offer resources for renewing entomological science.

Introduction

Our ability to understand mosquitoes is often clouded by the sense of discomfort their distinctive sound provokes – an auditory cue that signals itch, vexation, and even fear as potential vector of deadly diseases. Across human history, few sounds in nature have borne such a burden of hostility as this insect's high-pitched whine (Winegard 2019; Harrison 2025a; Renzi 2025c). Although scholarship has increasingly explored the sounds of other-than-human beings and their aesthetic, cultural, and symbolic dimensions, insects have largely remained at the margins of this discourse, despite their dense presence in acoustic niches and their significant symbolic roles within local soundworlds (Myers 1929; Dethier 1992; Medeiros Costa-Neto 2003; Rothenberg 2013; Carrier 2017; Manard and Coelho 2019; Feld 2025). Within the insect world itself, mosquitoes occupy an even more marginal position, with only a small and ambitious body of literature addressing their supposed musicality or even simply associating aesthetic values to their sonic distinctiveness (Morrison and Lemke 2022; Harrison 2025a, 2025b; Renzi 2025c; cf. Hawkes and Hopkins 2022, 21; Hoy 2024, 25). Yet, it is striking (if not ironic) that in entomology the very act that ensures the reproduction of this infamous dipteran – its mating ritual – is conceptualized in explicitly musical terms: the *dancing swarm*.

This article takes that notion as a point of departure and asks: how can mosquito sound and movement be taken seriously as aesthetic practices in their own right? And what kinds of listening practices are required to free our hearing from the encultured habits that relegate this insect's sound to the status of mere nuisance?

Whereas – through terms like "dance" or "song" – bioacoustics and entomological sciences have framed mosquito sound and movement in aestheticizing analogies that translate their phonokinetic behaviors into human-musical benchmarks, we turn instead to alternative sound ontologies and ethno-entomological perspectives, which acknowledge them as concrete lived realities, rooted in shared aesthetic dimensions, social existences, and multispecies kinships with humans and broader ecosystems. By adopting a biocultural approach that foregrounds situated, relational ways of knowing beyond nature-culture divides, the article attempts at positioning mosquito sound and movement within their own musical and choreutic dimension.

If we take distance from ethno-anthropocentric comparisons and mainstream epistemologies where the whining of mosquitoes persists as «one of the most recognizable and aggravating sounds on earth for 190 million years» (Winegard 2019, 7), and we rather tend our ears to systems of knowledge that – similarly to the insect – have been placed at the margin of global listening, we may begin to glimpse grounds for reimagining the mosquito not as a mere disturbance to swat away, but rather as a critical actor in shared soundworlds where its voice unsettles anthropocentric assumptions and renews science by expanding how we know, listen, coexist with *less desirable* organisms and within complex ecologies.

Arctic and sub-Arctic geographies provide a unique lens for engaging meaningfully with the mosquito, possibly because – to present – relatively few species in these regions act as dangerous disease vectors for humans or other animals (Koltz and Culler 2021; Culverwell and Vapalahti 2023). It is reasonable to infer that this relative absence of lethal threat has facilitated a more fertile space for dialogue and understanding with such a controversial organism since time immemorial (cf. Ford et al. 2022). Across these regions, a plurality of Indigenous traditions of multispecies storytelling has framed the mosquito as kin rather than foe, offering unprecedented grounds for science to listen to what their buzzing might reveal in a rapidly changing world – whether about their role in fragile ecosystems, the consequences of biodiversity loss, or, at times, even unexpected aesthetic dimensions of musical value. From the Inuit of Nunavik to the Haudenosaunee and Wôbanakiak in northeastern Turtle Island, from the Haida of the Pacific Northwest to the Ainu of northern Japan, Indigenous onto-epistemologies provide multiple angles on human–mosquito entanglements: through songs, stories, and dances that position the insect within shared origin narratives and land-based relationships, these traditions of knowledge transmit ecological understanding that resists dismissing mosquitoes as accidental or meaningless beings, and instead situates them as part of a relational and moral universe (Batchelor 1894; Simpson 2010; Sonawahes Antone 2018; Cherry 2019).

Among the many mosquito narratives circulating across the circumpolar North, this article focuses on their expression in multispecies storytelling from Sápmi, the ancestral and unceded homeland of Sámi peoples, spanning northern Norway, Sweden, Finland, and the Kola Peninsula in Russia. The most prominent form of such storytelling is *juoiggus* – hereinafter *joik* – a highly sophisticated vocal tradition that Sámi have shared for millennia with other-than-human beings inhabiting the tundra and Arctic coasts (Utsi 1998). Joik can thus be regarded as a biocultural heritage: it does not operate within nature–culture divides, foreign to Sámi ontologies, but instead inherits its notes from the sounds of birds, streams, wind, and – yes – even mosquitoes (Skaltje 2005, 12).

While ethnographic literature has generally dismissed non-human agency in the custodianship and transmission of joik – defining it instead as an exclusively human cultural practice – Sámi acoustemologies affirm that it cannot be confined to the human realm, and rather may also be voiced by any organic or inorganic feature of the land (Valkeapää 1985, 2001; Somby 1995). Sámi scholar and filmmaker Maj-Lis Skaltje (2005) has described joik as more-than-music (*eanet go musihkka*), a Sámi concept that captures the tradition's capacity to condense ways of knowing through sounding and listening: an Indigenous science that – among other things – enables interspecies communication and the transmission of ecological knowledge (cf. Renzi 2025a).

Building on this bottom-up definition, we sustain that joik, as multispecies and more-than-musical storytelling, offers a unique lens for engaging with mosquito sounds. By situating the insect as an active contributor to this biocultural heritage, Sámi understanding of joik unsettles anthropocentric analogies and assumptions about mosquito musicality and offers resources for renewing entomological science.

The extent to which academic research has been reproducing colonial approaches and systems of thought that reinforce objectification and systematic dismissal of local epistemes has been widely documented (Kuokkanen 2007; Kovach 2009; Simpson 2014; Robinson 2020). Against this backdrop, Sámi researchers and institutions have taken a leading role in asserting the importance of emancipation, co-production of knowledge, and the indigenization of the sharing process within academia (Kuokkanen 2000; Hirvonen 2008; Guttorm 2021; Virtanen et al. 2021). Our methodology is designed in continuity with these decolonizing horizons – appropriate and essential for a collaboration between Indigenous and non-Indigenous authors, and centered on Indigenous acoustemologies.

In what follows, we draw on the convergence of Renzi's research and field recording practice, focused on the joint consultation of bioacoustics and Sámi storytelling (2025c), and Utsi's performative and academic practice, both as a custodian and teacher of joik – particularly renowned for her voicing of the mosquito joik – and as a scholar of storytelling traditions and research storying in Sápmi (1998).

Joik and other forms of multispecies storytelling therefore serve in this article as subject matter and analytical lens, alongside field recordings and ecoacoustic analysis presented throughout the text and multimedia. Together, they define a relational approach to mosquito sound and movement in which local, lived, and performative ways of knowing frame and orient the scientific method. We treat joik and Sámi stories of mosquitoes as ethno-entomological data – valid scientific (situated) evidence of ecological knowledge rather than supplementary narratives – since they carry accountable epistemes while simultaneously signifying relationships and reminding us «who we are and of our belonging» (Kovach 2009, 94; cf. Utsi 1998; Guttorm 2011). But joik also epitomizes an Indigenous theory of knowledge and an exquisitely Sámi way of storying research – one that unfolds through attentive listening and sounding (Somby 1995; Renzi 2025a). As such, within the pages of this article, joik tightens the methodological gap between Indigenous sonic science and mainstream ecoacoustics. Utsi's mosquito joik combines rigorous ecological observation with ancestral understandings of environmental sounds, presenting us authors with a research methodology that keeps Indigenous paradigms audible in the foreground (responsibly guiding how listening and knowing are understood within Sámi communities) while also functioning as an evidentiary practice brought into dialogue with dominant scientific frameworks.

From here, the article unfolds as an alternate montage of stories – or rather histories of listening – that bring into dialogue distinct ways of knowing and hearing the mosquito. It begins with entomological and bioacoustic research, which quantifies wingbeat frequencies, swarm dynamics, and mosquito hearing sensitivity with remarkable precision, while at the same time gesturing toward a biocentric understanding of the insect's aesthetic behavior – an understanding still constrained by human-centered modes of listening. Our ears then turn to Sámi ecological knowledge (*árbediehtu*), which situates mosquitoes in kinship with herders and in multispecies alliances with reindeer, ants, and the tundra wind – as both ecological asset and vengeful torment. These perspectives acknowledge mosquito aesthetics as lived realities expressed through the relational practice of joik. In analyzing human–mosquito entanglements through joik, the insect emerges as custodian, performer, and teacher of the tradition, inviting us to take seriously the aesthetic and biocultural sociality only hinted at in entomology. The article concludes by bringing the discussion back to mainstream scientific frameworks: while technological advances produce increasingly precise data and pursue biocentric aims, these still fall short of addressing aesthetics satisfactorily. In light of what mosquito storytelling from Sápmi contributes to entomology, we leave open what a musical recognition of mosquitoes in biocultural terms might mean for science and for multispecies coexistence.

The dancing swarm: an entomologist's history of listening

Each summer, across the vast Arctic tundra, myriads of mosquitoes rise into dense, whining clouds above reindeer herds roaming in search of pastures. Entomologists describe these gatherings as gendered displays of courtship, expressed through patterned swarming around warm-blooded targets – a recurring behavior for mosquitoes across diverse habitats (Attanasi 2014; Becker et al. 2020; Baeshen 2022). Indeed, when mature enough to mate, mosquitoes engage in what Spielman and D'Antonio call a "dancing swarm": a ritual in which hundreds or even thousands of male *Culices* vociferously assemble above a specific landmark – known as a "swarm marker" – to attract hematophagous females (Spielman and D'Antonio 2001, 8). In the sparse tundra ecosystem, reindeer stand out as one of the few large, non-predatory warm-blooded hosts, making them ideal swarm markers for mosquitoes at the height of their reproductive frenzy. Reindeer migration also coincides with peak mosquito season, ensuring the insects can reliably locate the *Rangifer* each year, swirling in masses over the moving herd. This chronotopic entanglement provides a compelling lens for interpreting mosquito behavior as fundamentally organized around sound and coordinated movement.

Before it is heard as the distinctive and aggravating whine dreaded by humans (cf. Winegard 2019), the buzzing of a mosquito exhibits a remarkable ability of «exquisitely engineered organisms» (Hall and Tamïr 2022, 7). If we set aside, even briefly, the irritation their sound often provokes and instead attend to the almost imperceptible changes in pitch and flight path, mosquitoes appear as some of the world's most skilled listeners and accomplished sonic actors, with acoustic and aesthetic qualities that to date remain underappreciated. These dipterans are «endowed with one of the most subtle sound detectors in the insect kingdom, the Johnston's organ», a mechanosensitive structure apt at detecting even the faintest sound wave, allowing «a flying mosquito to detect the tones produced by the wingbeats of other flying mosquitoes and distinguish these from tones produced by its own wings» (Hawkes and Hopkins 2022, 21).

Since Pliny the Elder, humans have left records of their wonder at how such a tiny organism could generate such an outsized and penetrating sound (Pliny the Elder 1906[AD 77–79]; cf. Renzi 2025b, 2025c). The answer to this age-old question lies in the rapid, continuous rotation of their wings, which fends the air at a frequency that produces a relatively steady tone of around 300–600 cycles per second (Arthur et al. 2014). Particularly, the wingbeat frequency of male mosquitoes hovers around 600 Hz, while that of females typically ranges between 300 Hz and 550 Hz, with the lower frequency occurring when they are bigger or engorged (Spielman and D'Antonio 2001, 22; Becker et al. 2020, 21-22; cf. Arthur et al. 2014). In fact, the tonal quality of a mosquito's buzz is proportional to its size – smaller mosquitoes produce higher-pitched sounds, while larger ones, such as female mosquitoes, generate lower frequencies due to their larger wings, which beat fewer times per minute during flight (Spielman and D'Antonio 2001; Becker et al. 2020) (Fig. 1a).

When male and female mosquitoes mate, however, they match frequencies. And, more interestingly, they do so not at their fundamental wingbeat frequencies but at an extraordinary harmonic near 1200 Hz, a frequency that lies outside the typical auditory range of both sexes, as noted by neurobiologist and bioacoustician Ronald Hoy – who rhetorically asks in awe: «How in the world are they duetting where they can't be hearing?» (Cornell University 2009; cf. Hoy 2024, 25) (Fig. 1b). Mosquitoes' acoustic skills and sensitivities are thus determinant factors in swarm formation (Spielman and D'Antonio 2001, 23; Attanasi et al. 2014) and, particularly, for courtship rituals, during which the matching frequency of male and female wingbeats during flight creates «a harmonious duet in a prelude to mating» (Hawkes and Hopkins 2022, 21; see also Arthur et al. 2014; Menda et al. 2019; Hoy 2024, 25).

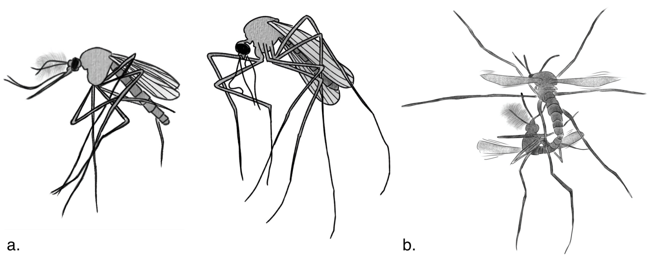

FIG 1. (a) Male mosquito (left) and female mosquito (right). (b) Mating pair in flight. Illustration by Astrid E. Hakvaag.

Descriptors like "harmonious duet" or "dancing swarm" curiously aestheticize mosquito buzzing as a musical object. Yet what is at stake in these comparisons is not a recognition of mosquitoes as aesthetic beings, but a translation of finely attuned phonokinetic behaviors into ethno-anthropocentric idioms. Within mainstream science, "music" and "dance" remain figures of comparison rather than evidenced aesthetic and performative realities for mosquitoes themselves. And in practice, this dominant epistemic orientation still recasts the "music" of mosquito buzzing to an undesirable disturbance, an obstacle to restful sleep, and an acoustic relegated to the periphery of aesthetic considerations – even when incorporated into musical works that center mosquito sounds, this is often derided (Morrison and Lemke 2022, 145). As Manard and Coelho show through their analysis of a wide-ranging collection of folk songs mentioning insects, references to dipterans are among the most abundant, yet unlike other arthropods, mosquitoes are most often portrayed in a negative light that stands in stark contrast to harmony and other forms of musicality (2019, 63). Even tracing back through historical accounts and literary traditions, the mosquito high-pitched whine ranks among the few environmental sounds most persistently maligned in Western imageries and representations of nature (Renzi 2025c).[1]

Sure, some of the acoustic properties that distinguish the mosquito – its fluctuating pitch contour, a timbre both coherent and rich in harmonics, and to some extent its droning textures – reveal an inherent musical potential. Yet these qualities are usually dismissed as nothing more than an incessant nuisance, especially when the sound assails us at night in our bedrooms. This dismissal reflects a Cartesian dualism in which the encultured habits shaping our hearing – what Kanngieser describes as "sonic colonialities" – are rooted in an ontology that draws sharp distinctions between culture and nature; between music, confined to the study of human cultural and aesthetic domains, and environmental sound, referred either to ambience or to the realm of ecological analysis (cf. Kanngieser 2023, 691-692).

Although sound- and music-related disciplines have moved in a transcultural direction where musicality is no longer used as a discriminatory label in the categorization and description of practices that deviate from dominant aesthetics (Titon 2015; Feld 2015; Giannattasio and Giuriati 2015; Staiti 2018; Ferlaino 2024), human musical expressions «remain, and will for some time, the fundamental benchmark for analyzing and evaluating other aesthetic sound communications from the animal kingdom» (Martinelli 2011, 141; cf. Lewy 2017). In this respect, zoomusicologist Dario Martinelli provokingly asks

> what rational explanation there could be for *denying* that animal species (and *Homo sapiens* is one of them, as we all know) have aesthetic capacities and sensibility? If aesthetics is, as it is, a biological phenomenon, then it is just obvious that such processes have an active role in animals' cognition. (Martinelli in Ullrich 2014, 117)

Martinelli advocates a biocentric approach that aspires to emic directions and starts from species in which an aesthetic action (creation/performance) elicits an aesthetic reaction (attraction/repulsion). He exemplifies this with the case of male humpback whales, who develop and perform complex songs to

[1] In this article, the term "Western" refers primarily to dominant epistemological frameworks that emerged in Europe and were later consolidated through colonial powers also in North America and Australasia, especially within scientific rationalism and colonial governance. As noted in Indigenous research methodologies, "Western" and "the West" are contested and relational terms, often defined in opposition to what is framed as Indigenous. For this reason, we also use "dominant", "hegemonic", or "mainstream" to describe epistemic traditions of European origin that historically set norms while marginalizing other ways of knowing (Wilson 2008; Virtanen et al. 2021).

compete with other males, while females are drawn to the most successful performance (ibidem; see also Martinelli 2011, 31). A similar approach could be applied comparatively to a biocentric inquiry of how musical mosquito is since, as previously noted, the action/reaction dynamic described by Martinelli is also observable in the dancing swarm ritual. This is precisely what Ronald Hoy sought to demonstrate in his experiments with the *Aedes aegypti* in the laboratories of Cornell University. Hoy argued that acoustics plays a crucial role in mosquito mating behavior and carried physiological recordings directly from the Johnston's organ itself to observe its responses to distinct wingbeat frequencies. The results showed that females are attracted (reaction) by the tone produced by a male partner (action), with whom she ends up "duetting" by adjusting the wingbeat to come together at a common tone:

> What are they doing with almost as many sensory auditory cells as we have? Maybe they have to be tuning themselves very finely in frequency, just as we do. And wouldn't that be an interesting convergence, mosquitoes and people? The next time you hear a mosquito buzzing around in your ears, you might be thinking about the fact that there's a lot more going on in this buzzing than just flying over to give you a nibble. In fact, that buzzing sound is a mating call for many mosquitoes. And that mating call isn't a simple buzz. There's a lot going on, as far as a love duet between male and female. They are interacting on the fly and in song. (Ronald Hoy in Cornell University 2009)[2]

If a fully "emic" biocentric understanding of other-than-human aesthetic behaviors may however still seem out of reach as an alternative to anthropocentric comparative models, how can we truly approach mosquitoes musically and choreographically on their own terms? How can we grasp mosquito sound ontologies and aesthetics without resorting to metaphor? Looking across the plurality of knowledge systems – beyond a positivist science, whose observational approach draws sharp lines between humans and nature, and as a result between music and environmental sound – intermediate biocultural pathways guided by relational sound ontologies and acoustemologies not bound to the nature-culture divide can offer alternative teachings and models. Such pathways – frequent among Indigenous epistemes – open the possibility for a broader, more meaningful and more serious inclusion of diverse voices in the debate on musicality, even those that we conventionally regard as alien or undesirable (Skaltje 2005; cf. Feld 1988; Martinelli 2011; Brabec de Mori 2013; Lewy 2017). They also support discernment of multispecies entanglements and historicities in which mosquitoes are meaningfully embedded within human and more-than-human societies.

The "interesting convergence" between mosquitoes and people that Hoy has long wondered about can be found in these underrepresented ontologies that, because of their relational nature, offer fertile ground for investigating and acknowledging shared aesthetics and musical biocultures – cases that may help us approximate just *how musical* mosquito is. And so we turn to Sámi histories of listening, where ecological knowledge and multispecies kinships with mosquitoes are carried through oral performances, landscape practices, and deep listening with the land.

The herding swarm: a Sámi herder's history of listening

Before a certain time in the history of the European Arctic, mosquitoes were absent from Sápmi. This absence has long raised local questions: where did this insect come from, and why did it move to the ancestral land of the Sámi? What does it signify in the North? And why is it important that some of them are referred to as *she* – as mosquito "girls"? Stories about their origins have been passed down orally among the Sámi since time immemorial – verbally or through joik – until 1920, when the pioneers of Sámi literature Johan and Per Turi recorded them in full in the anthology *Lappish Texts* ("About the mosquitoes in olden times").

The story tells how mosquitoes first arrived in Sápmi after a long journey from southern Finland, led there by a spider. The spider spoke of a place – or rather a land – far up in the North, rich with fat reindeer, abundant fish, and well-kept riverbanks and lakeshores. Tempted by these encouraging promises, the Mother of Mosquitoes selected three mosquito brothers (*golbma vieljaža*) to embark on an expedition to scout this dreamland. Along the way, however, each of the brothers met a different fate and never returned: one drowned in a milk trough, another became trapped in a tar bucket, and the third was suffocated by smoke inside a *lávvu*, the traditional Sámi conical tent. When the brothers did not come back, the mosquitoes who had stayed behind grew worried. The Mother of Mosquitoes consulted the spider, but he could not explain. Despite the concern, the rest of

[2] Transcript from the video "Mosquito Hearing", published by the College of Agriculture and Life Sciences (CALS) at Cornell University, on March 9, 2009: https://www.cornell.edu/video/mosquito-hearing#:~:text=Male%20mosquitoes%20have%20a%20wing,a%20harmonic%20near%201200%20Hz.

the mosquitoes set out northward in search of their lost kin. Upon reaching the North, the colossal swarm of bloodthirsty insects encountered a reindeer herd in the dense boreal forest. Never before had they seen reindeer – or such a multitude of warm-blooded creatures. The swarm rushed upon the herd and whirled around it with delight, their loud buzzing hurrying the exasperated animals up the mountain. From that moment, once the mosquitoes had reached Sápmi, they never left (Turi and Turi 1920, 124; see also Palomaki 1993, 5; Utsi 2025).

The story of how mosquitoes came to Sápmi is thus intimately connected with the movements of reindeer. More than a "cosmogonic" myth, this narrative conveys deep Indigenous science (*árbediehtu*, in North Sámi), through which the Sámi not only learn to shield themselves from mosquitoes – by making use of the same substances that once killed the *golbma vieljaža*, such as tar and birch smoke – but also acknowledge the mosquito as a powerful force capable of setting entire herds in motion along their traditional trails across the fjells. This multispecies entanglement reappears in Johan Turi's writings. When describing the ethology of the reindeer, he offered a vivid account of the mosquito's agency:

> Go šaddá báhkka, de bohccot ruhttet dakkár alla jieŋaid nala, gosa ii olmmoš beasa, ja ruhttet čuoikkaid (Turi 1910, 5)[3]

Such interdependency endures today in the traditional knowledge of Sámi reindeer herders as a gift that facilitates the assembling and movement of the herd. Ante Aikio, from the Finnish side of Sápmi, notes that in late June the insects, thirsty for reindeer blood, drive the animals into compact flocks until they grow in their thicker winter coats, which mosquitoes and horseflies can no longer pierce (cf. Müller-Wille et al. 2006, 33; Olsén et al. 2019, 174). It is during this time that

> [r]eindeer are so exasperated by the mosquito that they escape them trotting up the fjells where it's windier, and by moving towards the coast, where the wind is also stronger, and of course because of pastures. (Ante Aikio, oral teaching, June 2, 2021, Levi, Sápmi)

One reason reindeer gather into large herds when pressured by mosquitoes is that the collective airflow created by their breathing is stronger and helps keep the insects hovering slightly higher (Eira 2011; Olsén et al. 2019). A second reason surfaces in another key oral narrative centered on mosquitoes, transmitted this time in the form of a *dajahus* – the poetic content of a joik (Video 1):

/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.1) Ja mihcamáraide jo guovtti soabbái	And at Midsummer on two sticks
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.2) Ja sápmelačča buoremus jo biigá	The best maid of the Sámi
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.3) Ja bieggabárdni lei jo láidesteaddji	And *bieggabárdni* was the leader
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.4) Ja čuoikanieida lei jo vuojeheaddji	And *čuoikanieida* was the driving force
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.5) Ja golbma vieljaža jo ledje,	The three mosquito brothers
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.6) Ja okta heavvanii jo mielkegivlui,	One gets stuck in a tar bucket
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.7) Ja nubbi darvánii jo bihkkalihttái	One drowns in the milk trough
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…
(v.8) Ja goalmmát hávkkai suova sisa jo,	One suffocates by smoke
/: Na lo, le, loi, lei, loi-le, lo :/	Na lo, le, loi, lei, loi-le, lo…

Vid 1. Mosquito joik performed by Mai Britt Utsi. Excerpt from Utsi 2025. youtu.be/wdtICqlKK1k

While this *dajahus* contains few words, each is exquisitely crafted to exalt the mosquito's agency and significance within the relational ecologies of the tundra – ecologies through which the Sámi continue to celebrate, appreciate, and remember this insect. Certain verses directly reference the origin story of the mosquito in Sápmi, particularly alluding to the disappearance of the *golbma vieljaža*, while the opening line situates the listener in the precise season when mosquitoes are at their most ravenous – Midsummer.[4] Here, the metaphor of "two sticks" evokes the red, swollen marks left by mosquitoes on a victim's skin, «just as hikers leave marks on the ground or snow when walking with their two walking sticks» (Johan Sara Jr., oral teaching, April 29, 2021, Máze, Sápmi). At this time of year, it is easier to gather the reindeer into a fence or corral for calf marking, since

[3] In the translated edition: «When it gets hot, the reindeer head up onto high glaciers where people cannot follow, and because of the mosquitoes» (Turi 2011, 14)

[4] In Sápmi, the ubiquity and ecological relevance of this small dipteran (particularly, *Culex Pipiens*, *Culex Pulicaris* and *Culex Reptans*) in the summer is so significant that it has been linguistically assimilated into weather terminology. For example, the word *čuoikabalvái* refers to a specific weather condition characterized by clouds of *Culicidae* and other Arctic flies (*Diptera*: *Simuliidae*) that fill the air and even obscure the sky. Due to the seasonality of this phenomenon, mosquitoes lend their name to an entire time of year: *čuikit*, when the weather begins to be such warm and pleasant that mosquitoes, gnats, and other flies become especially abundant (Nielsen 1932, 423, 433; Eira 2011, 41).

they may be found already naturally assembled in the highlands as one herd. Verses 2 to 4 of the *dajahus* precisely convey this knowledge, presenting the second reason why reindeer cluster into large herds under mosquito pressure. In the second verse, the insect is described using a specific kinship term, *biiga*, a North Sámi word that translates as "maid" or "helper". Within the herding unit, this role refers to a young person whose tasks include, among other things, helping to round up and care for the herd:

> Dajaldat "čuoika lea badjealbmá reaŋga" doallá deaivása, go čuikenáigi ja čuoikkaid mearri váikkuhit hui ollu maid bohccuid geasseeatnamiidda johttámii. (Olsén et al. 2019, 150)[5]

Tormented by mosquitoes in the woods, the reindeer seek refuge in the wind, gathering as a full herd on the summits of the tundra fjells after being rounded up by the tiny yet relentless insect. III–IV verses describe the very forces at work in this herding process – one that strikingly echoes the gendered dynamics that entomologists identified as the "dancing swarm". In fact, the verses sequentially describe the *choreographic* complicity between the male mosquito (*bieggabárdni*) and the female mosquito (*čuoikanieida*).

While *čuoikanieida* simply translates as "mosquito girl", *bieggabárdni* is a compound word meaning "son of the wind" and relates to a broader multispecies kinship terminology that parallelly labels the reindeer as the asset of the wind (*bieggabuorre*) before even being an other-than-human animal herded by humans (Sara 2001). Much like the wind, whose direction serves as an essential compass for the reindeer (ibidem; cf. Renzi 2024), the male mosquito – the wind's son – guides and orients the herd's movement (as *láidesteaddji*). It is for this affinity, and for his remarkable ability to do so, that he earned the name *bieggabárdni*. The male mosquito moves to the front of the scattered herd, teasing the lead reindeer with his high-pitched buzzing and swift swirls until the animal trots in search of wind, pulling the others along.

The orchestration of the herd, however, is said to be choreographed by the strongest and loudest of the mosquitoes, *čuoikanieida*, who is the best Sámi *biiga* (cf. Utsi 2025). *Čuoikanieida* – the she-mosquito – is drawn to the abundance of succulent, warm-blooded prey revealed by the strident buzzing and outstanding herding display of *bieggabárdni*. While *bieggabárdni* gathers and pulls the herd from the front, the female provides the driving force from behind (as *vuojeheaddji*), pressing the reindeer forward so that they climb toward the windy ridges, where the pastures are good, the insects fewer, and patches of *jasát* (snowspots) still cool the summer heat (Sara 2001). Through their coordinated sounding and movement, male and female mosquitoes round up and drive the herd (Fig. 2).

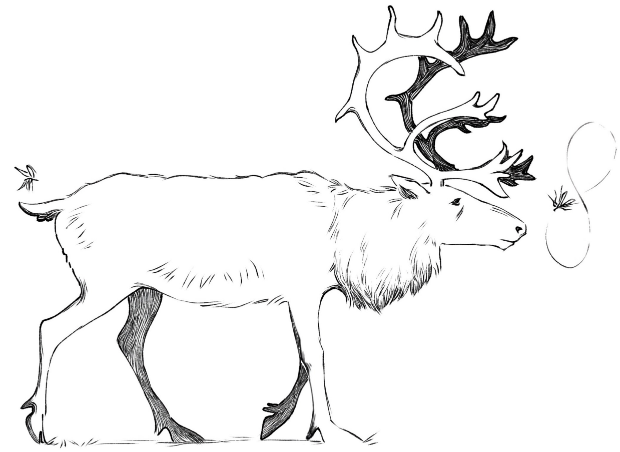

FIG 2. Illustration of the dancing swarm with Sámi ecological knowledge. *Bieggabárdni* (swirling out front) and *čuoikanieida* (stinging from behind) team up to tease and herd a reindeer 'swarm marker'. Illustration by Astrid E. Hakvaag.

The joiking swarm: an Arctic mosquito's history of listening

Across centuries, foreign explorers and naturalists have described the relationship between mosquitoes, people and reindeer in the Arctic as a sort of interspecies torment (Renzi 2025a, 2025b). By contrast, as the previous paragraph has shown, Sámi epistemes – imbued with biocultural values and expressed through multispecies storytelling – tend to locally frame mosquito buzzing as a powerful voice that sets tundra ecologies in motion and links the insect to herding practices and interspecies kinships. Yet it would be an oversimplification to suggest that the Sámi never experience mosquitoes and their sound as torment.

In the 1960s, a Sámi couple from Várjjat (Varanger, Finnmark) traveled by car across Sápmi to the Anár (Inari) municipality in northern Finland. After visiting the local museums and shops, they found a dry, suitable place to spend the night. There they built a campfire, sat on reindeer skins, and shared a simple Sámi meal of roasted reindeer meat, followed by coffee, cloudberries, and cream. As they sat by the fire, the man noticed a nearby anthill and began to tease the ants with a stick. The ants struggled to defend their home, but he brushed them away one by one. His wife urged him to stop and warned

[5] "The saying 'the mosquito is the reindeer herder's helper' is therefore accurate, as the timing of the mosquito's appearance and their numbers significantly affect the reindeer's movement toward summer grazing areas."

that the ants would eventually take their revenge. He dismissed her concern, saying that since they would be sleeping in the car, they could rest undisturbed, as the vehicle would prevent any intrusion. When he finally abandoned that "amusement", the couple chatted for a while in the glow of the midnight sun before moving into the car to sleep. The woman rested soundly through the night. The man, however, was repeatedly awakened by a lone mosquito that had separated from the swarm and managed to infiltrate the vehicle. Each time he drifted off, the insect returned, joiking loudly in his ears and keeping him from resting. He knew well that killing it would only invite more – *if you kill one, ten will take her place*, goes the saying – making the joiking even louder. The following morning, the woman noticed her husband's exhausted face, and he recounted the harsh night he had been through. She reminded him of her earlier warning: this was the ants' revenge. Even if they could not enter the car themselves, all beings on earth are connected, and even the smallest creatures communicate and collaborate. When they returned to Várjjat, the man repeated the story to his children, warning them never to disturb the ants – for otherwise mosquitoes might keep them awake all night with their loud and incessant joik.

This oral teaching seems to situate the mosquito as a nuisance in ways similar to the most globally shared perceptions of the insect. Yet it is striking that the buzzing which torments the man through the night is described as "joiking" – that in Sápmi carries strong positive aesthetic and affective associations with the beauty, vitality, and voice of the land. Naturally, this should not be interpreted as suggesting that the joik itself is the source of torment – in the same way that early European travelers once dehumanized and dismissed the tradition as lacking musical value (Hirvonen 2008, 133).[6] Rather, it gestures to the paradox that even a cherished song, when untimely, may weigh heavily on the listener's ear. Nor should it be read as implying that mosquitoes are poor joik performers or that their joiking is unappreciated – on the contrary, several oral accounts recall Sámi hearing and learning many joiks precisely by listening to mosquitoes.

The recognition of mosquitoes as skilled joik performers – beyond metaphoric language – is deeply embedded in Sámi acoustemologies and explicitly surfaces in the traditional *dajahus* and storytelling associated with the insect. In 1868, Swedish proto-ethnographers Gustaf and Lotta von Düben collected several *dajahus* alongside their anatomical survey of the Sámi in northern Sweden. Among these is a mosquito joik, transcribed in Swedish by the South Sámi priest Anders Fjellner (1795–1876) from Härjedaelie (Härjedalen). In the *dajahus*, a grasshopper interrogates a mosquito about what she does during the summer. The mosquito replies that she sings ("Jag sjunger", in Fjellner's transcription) and then poses the same question back to the grasshopper, who answers that she dances to the music created by the mosquito.[7] Although no contemporary version of this *dajahus* is preserved in South Sámi, it is reasonable to infer that – in the original version – the verb translated in Swedish as *sjunger* ("to sing") was in fact the South Sámi verb for the act of joiking (*joejkem*), as the distinction between joiking and singing traditions only became marked under external musical influence in Sápmi (cf. Jones-Bamman 1993). This reading seems to be supported by current practices. Sámi joiker and jurist Ánde Somby shares a personal story of how he came to learn a mosquito joik, highlighting a process grounded in prolonged, attentive listening, and in a broader – necessarily patient – embodied experience.

FIG 3. Ánde Somby joiking with mosquitoes. Frame from film *Advocate of Yoik* (© Renzi 2025)

To learn the joik from the insect, Ánde refrained from killing any mosquito for years, allowing entire swarms to «joik into [his] ear and willingly getting stung» to experience both their acoustic and tactile presence (Ánde Somby, oral teaching, May 19, 2021, Sirbmá, Sápmi; cf. Aubinet 2020a, 186) (Fig. 3). This embodied experience translates into his performance, where Ánde Somby's voice seeks to achieve

[6] The most notorious passage reads: «The juoige, or song of incantation, is used by the Noaaid whilst in the exercise of his magical function. To say it is sung, is to give an imperfect idea of the magician's manner of delivering it, which he does in the most hideous kind of yelling that can be conceived» (Acerbi 1802, 311).

[7] The Swedish transcription recites: «Gräshoppan frågar myggan: "Hvad gör du om sommaren?" "Jag sjunger", svarar myggan; "men hvad gör du sjelf?" – "Jo", svarar gräshoppan, "jag dansar till din musik"» (von Düben 1873, 346).

phonomorphism through a skilled use of falsetto that blends with the high-pitched tone of the mosquito. This process of *becoming* is further accomplished through a largely stable melodic profile centered on a single note – exception made for brief excursions and embellishments that do not exceed a sixth from the central tone – which allows the *luohti* to reach the audience as the recognizable undulating yet obstinate pitch that a mosquito emits when flying close to one's ear (Fig. 4).

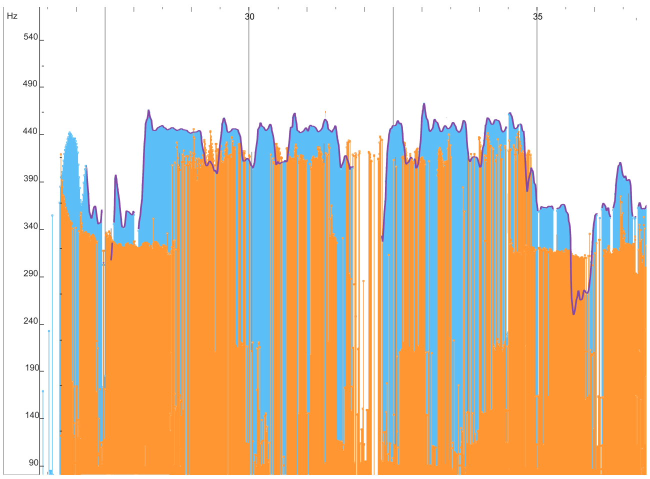

FIG 4. Comparison between the stem plot of the estimated fundamental frequency from Ánde Somby's mosquito joik (in blue; contour highlighted in purple), and from an individual mosquito's wingbeat (in orange) recorded through the 'cup method' (cf. Sinka et al. 2021).

By reaching the listener's ears as a mosquito, Ánde Somby can convey the insect's message through a *dajahus* which

> establishes the existential logic of the mosquito within nature. Through her own voice, the mosquito signals exactly when she is arriving and what she is going to do. And by doing so, she says:
>
> *I sting you.*
> *I will sting you because blood is so sweet.*
> *I sting you.*
> *I sting you because it is through this little sting*
> *that you will notice you are alive.*
> *And the day crows will turn white,*
> *I will be done with my task:*
> *To guarantee the continuance of life.*
>
> This way, the mosquito can teach us about the ecological role they play on the stage of existence. (Ánde Somby, oral teaching, May 19, 2021, Sirbmá, Sápmi)

The *dajahus* is particularly relevant to this discussion. It underscores not only the recognition of the mosquito as integral to the dynamics of tundra ecosystems – similarly to the *dajhaus* of *čuoikanieida* and *bieggabárdni* for reindeer herding – but also how the "annoying" qualities of the insect are inseparable from the ecological benefits it provides and from the acknowledgment of its aesthetic place within the joik heritage. To advocate a shift in perspectives on the widely conflictual human-mosquito relationship, Ánde Somby embraces the very *perspectivism* intrinsic to what he calls «the shamanistic view» (*noaidevuohta*) of joik (ibidem). Rather than knowing the mosquito through positivist objectification or mythic discourse, *noaidevuohta* presupposes a mode of knowing through becoming, where joik allows humans to embody the perspective of the other-than-human being sought for understanding (Somby 1995; Helander-Renvall 2010, 50; Porsanger 2012, 42; Aubinet 2020a, 117):

> As a human being, I'm bound to this body and I perceive the world as such. But, through joiking, I get to perceive the world as a mosquito to understand why she behaves like she does. And through joiking I can also be perceived by other humans as a mosquito, conveying her messages. (Ánde Somby, oral teaching, May 19, 2021, Sirbmá, Sápmi)

A listener fluent in North Sámi would in fact discern an additional layer within Ánde Somby's zoomorphic performance, as its *dajahus* is delivered in the first person, from the mosquito's perspective, facilitating communication by the insect through a human body and language (cf. Aubinet 2020a, 117) (Video 2).

VID 2. Mosquito joik performed by Ánde Somby during a concert organized by Nicola Renzi at the Fondazione Giorgio Cini, Venice: https://www.youtube.com/watch?v=NI4Dt-n9TJ6U&t=5s (starts at 3'27''). Filmed by the author.

It is undeniable that a defined melodic contour can be detected in Ánde Somby's voice as he brings the mosquito into being through joik. Skilled vocal technique is fundamental to the Sámi ability to joik a subject into presence, and to the successful rendition of an animal joik more broadly (Aubinet 2020b). With specific reference to the mosquito joik, Inger Julianne Eira underscores how

> [m]ohkiid, ravggodemiid ja fanahallama bokte sáhttá ovtta ládje gullat ahte lea čuoikkanieidda luohti. Go diehtá ahte lea dan luohti, de easkka sáhttá govahallat ahte livčče dego čuoika mii boahtá bealljemáddagii njurgut, go juoigi mohkasta ja alida jiena. Vaikke lea čuoikka luohti, beassá

vuostáváldi oahpásnuvvat makkár doaibma čuoikkas lea sápmelaččaid ektui.[8] (Eira 2022, 22)

The performer's vocal abilities make the mosquito vividly present to the listener's ear through melody. Yet, given the semiotic qualities of this vocal tradition – where signifier and referent are inseparable in a joik (cf. Somby 1995; Gaski 1999) – what we hear in the joiker's voice is not merely a melody, but the mosquito itself (Utsi 2025). This raises a further question: if the mosquito can be heard in a joik voiced by humans, can we also discern a joik within the *voice* of the mosquito?

One might ask whether the choice to describe the mosquito's buzzing as a joik – both in the revenge of the ants, in Fjellner's *dajahus*, and in Ánde's experience – belongs to the same rhetorical register as the terms "dancing" and "duetting" employed by entomologists. Yet Sámi joik ontologies frame the tradition as transcending human exceptionalism, characterizing it as a biocultural heritage shared among the Sámi and other Earth beings, where joik inherited its notes from the voice of wind, birds, rivers, and even that of mosquitoes (Skaltje 2005; cf. Renzi 2025a). Within this horizon, the mosquito stands – alongside any human and other-than-human subject in Sápmi – as custodian, performer, inspiration, and teacher of the joik heritage. From this relational ontology, mainstream sciences may find resources to move beyond Cartesian ways of thinking aesthetic realities, taking up the challenge posed in the introduction: to hear mosquitoes musically on their own terms and to engage with their aesthetics without reducing them to metaphor – something a biocultural heritage like joik has long practiced through lived, relational forms of musicking.

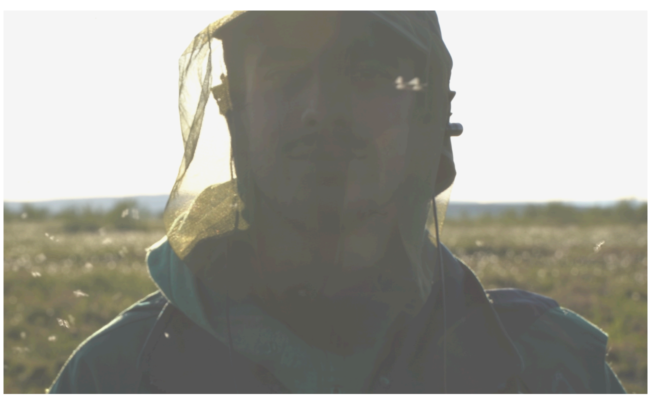

Fig 5. Recording mosquito joiks with binaural microphones at Govddaluovttafávli, Guovdageaidnu, 9 July 2023. Photograph of/by the author (© Renzi 2023)

As we authors discussed the materials on which to base this article and how to present a joik voiced by mosquitoes, we listened back to some binaural recordings collected (painfully) in vast bogs under siege from mosquito squadrons (Fig. 5). With omnidirectional capsules in the ears, one of us became a swarm marker around which the insects performed their dance. Just as their sound and movement compacts reindeer into dense herds, here the swarm gathered us together in the same room, listening intently to their acoustics. As the recording played, one of us instinctively scratched the ears in remembered irritation, while the other – with ears better attuned to that place – wondered aloud: do you hear a melody in there?

Čuoikas lea iežas jietna, jus don guldalat don sahttát gullat ollu luođit doppe, jus don veallát olggon, lavus ja gullat rággasis siste, don veallán gulla olggon bealde:

Ahte dies bealje maddagii jus dus ii leat rákkas! Ja de don sahttát álgit jurdašit:

Ii go lea duohta, dat lea vehá somá dat čuoika.

Each mosquito has its own voice, and if we listen carefully, we can hear many joiks within it. Lying outdoors – or even on the safer side of the inner tent in a *lávvu* – you hear the swarm approaching from outside. And then it begins to enter your ears too, especially if you do not have an inner tent. That is when you begin to hear it, to meditate. There, the joik comes. This is the beauty of the joiking swarm (Video 3).

Vid 3. "Do you hear a melody when you hear a mosquito?" Filmed by the authors. https://youtu.be/yV6rTTv4pl4

Conclusions

The technological advancement has provided increasingly refined tools to investigate how mosquitoes perceive and interact with their environment (Cornell University 2009). Yet a comprehensive understanding

[8] "Through swings, stretches and other vocal techniques, one can hear in one way that it is the *luohti* of the *čuoikkanieida*. When you know that it is that *luohti*, you can imagine it like a mosquito that comes whistling into the ear root, when the *juoigi* bends the voice and raises the pitch. Although it is the mosquito *luohti*, the listener gets to know the function of the *čuoikka* in relation to the Sámi people."

of their perspective remains elusive (Martinelli 2010, 83). As for houseflies, cockroaches, ticks, and other arthropods, our ability to come to terms with mosquitoes is obfuscated by a prevailing ontology that relegates them to what Hohti and MacLure describe as «the margins between life and death, organic and inorganic, human and nonhuman, conscious and unconscious» (2022, 328; cf. Ford et al., 123). Our capacity to truly hear mosquitoes is further deafened by an epistemology that predominantly links their sensory presence to irritation and fear, due to their sting's potential lethality. Within this "sticky" frame, their high-pitched whine has come to describe an acoustemology of nuisance: a disturbance to quietude, an obstacle to sleep, pushed to the periphery of aesthetic consideration, even when incorporated into creative works (David Rothenberg, personal communication, August 29, 2024, online; cf. Rothenberg 2013; Hawkes and Hopkins 2022, 20; Morrison and Lemke 2022, 145). This rather inescapable acoustemology has formed within a dominant history of listening where, over millennia, these dipterans have become synonymous with one of the most recognizable and undesirable non-anthropic sounds in nature (cf. Winegard 2019, 7; Renzi 2025c).

Entomological research has nonetheless hinted at the possibility that mosquito acoustic behavior can be meaningfully understood in aesthetic terms. Dynamics of attraction and repulsion through sound – recognized in other species as distinctive modes of multispecies musicking – also manifest in the courtship ritual of the dancing swarm. Yet when applied to mosquitoes, musicality as a benchmark for evaluating their aural aesthetics and communication is often treated as mere analogy rather than acknowledged as a lived reality.

In an era marked by severe environmental challenges and the erosion of ontological and epistemic diversity, the heterogeneous ways humans engage with other species through sound and movement offer fertile ground for complementing the ecological insights made possible by technological advancement in Western science. Indigenous sciences – here exemplified by Sámi *árbediehtu* and multispecies storytelling – articulate mosquito aesthetics not as metaphor but as lived relations, grounded in planetary kinship, shared sociality and heritages (such as reindeer herding and the joik tradition). They stand as rigorous, place-attuned systems of knowledge, carried and renewed in stories that are locally meaningful, while also capable of dialoguing with, expanding, and at times unsettling the understandings generated by mainstream entomology (cf. Kuokkanen 2000; Kovach 2009).

It has been noted how the ability of mosquitoes to round up reindeer into compact herds is of an exquisitely acoustic nature. The acoustic dimension of this herding act beyond the human is long known and passed down within Sámi acoustemologies – particularly through joik and other storytelling traditions. The lexical distinctions used to denote the gender identities and roles of mosquitoes (*bieggabárdni* and *čuoikanieida*) well illustrate a deep phenomenological knowledge of swarm ecology, where male and female mosquitoes work together as the best Sámi *biiga*. Herders recognize the gendered sonic dynamics that model the dancing swarm. Although they neither possessed the tools nor the interest to measure the exact frequencies at which *bieggabárdni* teases a reindeer "swarm marker" or aligns his wingbeat with *čuoikanieida*, Sámi *árbediehtu* – through joiking – has long captured their joint choreography eloquently. In this way, mosquito behavior is made sense of not only as mating ritual, but also as part of wider networks of ecological entanglement and multispecies kinship that bind insects, reindeer, wind, and human herders.

Listened through this biocultural lens, Sámi acoustemology of the mosquito can help approximate a better aesthetic understanding of mosquitoes in their own terms. The story of *čuoikanieida* and *bieggabárdni* is exemplary, as it acknowledges the choreutic dynamics of Spielman and D'Antonio's "dancing swarm" and resonates with Hoy's biophonic "duet", in which male and female mosquitoes converge at a common tone in their voracious choreography around reindeer. Yet *árbediehtu* also moves beyond these frameworks by situating the mosquito as a vital participant in the joik heritage. Here, the mosquito is not a metaphoric "dancer" or "singer", but a custodian, performer, and teacher of this relational practice. One with an equal claim to joik as any other being in Sápmi. One whose joiking may arrive as inspiring communication with human performers, or as torment when multispecies alliances are unsettled – such is the case of the "revenge of the ants".

The growing recognition of other-than-human beings as musical actors within anthropological and ecological disciplines points toward new horizons, but ones that still lack effective analytical methods capable of moving beyond cognitive frameworks constrained by nature–culture binomials (Martinelli 2011). While a fully biocentric analysis of how musical the mosquito is on its own terms may remain out of reach, biocultural approaches to its *voice* offer one way forward: interpretive tools less constrained by colonial epistemes that define music as exclusively human and cultural, and thus separate from nature and environmental sound (Kanngieser 2023, 692). The histories of listening with mosquitoes condensed in this article – of gathering reindeer, of joiking loud vengeance on behalf of the ants, of passing down beautiful melodies to human fellows – suggest how even the most controversial beings can reshape multispecies relations and recalibrate our listening as human beings. What may be embraced as a dear companion by some, and dismissed as an insignificant nuisance by others, ultimately depends on the posture we adopt toward these pesky insects – a posture ultimately determines our responsibility towards the biocultural sustainability of human-mosquito entanglements and broader ecosystems (cf. Hohti and MacLure 2022, 327).

By asking non-Indigenous listeners whether melodies can be heard in the buzzing of mosquitoes, hesitations often arise in unfamiliar ears. Do mosquitoes perform the same melody that the Sámi associate with this insect when they joik it – or perhaps other, known or yet-to-be-discovered joiks? And if so, which joik can people hear and learn in the musical *and* more-than-musical sound of mosquitoes? Such questions point to the need for further collaborative research with both human and other-than-human communities in Sápmi, with mosquitoes well exemplifying the very nature of these hesitations. In its multispecies storytelling – as a form of knowledge production that seeks to listen and stay "with the trouble" (Haraway 2016) – the mosquito joik and the joiking of the mosquito offer only a partial response, yet one that opens a glimpse into an alternative way of listening and thinking about mosquitoes. The possibility of an ecological and aesthetic convergence between mosquitoes and people, provided that we all attempt to actively *listen with the trouble* and decipher the messages these insects may convey.

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